Virtual University	Virtual Systems and Services Practical (CS505P) Assignment No. 01	Total marks = 20 Deadline Date: Nov 06, 2024

Please carefully read the following instructions before attempting assignment.

RULES FOR MARKING

It should be clear that your assignment would not get any credit if:

- The assignment is submitted after the due date.
- The submitted assignment does not open or file is corrupt.
- Strict action will be taken if submitted solution is copied from any other student or from the internet.

You should concern the recommended books to clarify your concepts as handouts are not sufficient.
You are supposed to submit your assignment in .doc or docx format.
Any other formats like scan images, PDF, zip, rar, ppt and bmp etc. will not be accepted.
Labs covered: 1- 5

Topics covered:

- Software-Defined Data Center (SDDC)
- Virtualization.

NOTE

No assignment will be accepted *after the due date via email in any case* (whether it is the case of load shedding or internet malfunctioning). Hence refrain from uploading assignment in the last hour of the deadline. It is recommended to upload your solution file at least two days before its closing date.
If you find any mistake or confusion in the assignment (Question statement), please contact your instructor before the deadline. After the deadline, no queries will be entertained in this regard.
For any query, feel free to email at cs505P@vu.edu.pk

Problem statement

Q1. Define virtualization and explain its primary types. How does virtualization benefit modern IT infrastructure?

Virtualization is the process of creating a virtual instance of computer hardware, storage, network, or other resources to allow multiple operating systems and applications to run on a single physical machine. By abstracting these physical resources into virtual instances, virtualization maximizes resource utilization, improves scalability, and enhances the flexibility of IT operations.

Primary Types of Virtualization:

Server Virtualization:

Definition: Divides a single physical server into multiple virtual servers.

Benefit: Improves server utilization, reduces hardware costs, and simplifies server management.

Storage Virtualization:

Definition: Pools multiple physical storage devices into a single, virtual storage unit.

Benefit: Simplifies storage management, improves performance, and allows dynamic allocation of storage resources.

Network Virtualization:

Definition: Combines multiple network resources and manages them as a single, software-based virtual network.

Benefit: Enhances network agility, optimizes bandwidth usage, and improves network scalability.

Desktop Virtualization:

Definition: Allows users to access their desktop environment remotely from various devices.

Benefit: Facilitates remote work, increases security, and reduces IT maintenance for individual devices.

Application Virtualization:

Definition: Separates applications from the underlying operating system, enabling applications to run in isolated environments.

Benefit: Simplifies software updates, enhances application compatibility, and improves security.

Benefits of Virtualization in Modern IT Infrastructure:

Cost Reduction: Reduces the need for physical hardware, saving on purchasing, maintenance, and energy costs.

Improved Resource Utilization: Increases resource utilization by allowing multiple virtual machines to share the same physical hardware.

Scalability and Flexibility: Facilitates rapid scaling and deployment of applications, allowing businesses to adapt to changing demands.

Disaster Recovery: Enables efficient backup and recovery of virtual machines, ensuring minimal downtime during disruptions.

Enhanced Security: Isolates applications and data, limiting the spread of malware and security breaches.

Q2. Discuss the benefits and challenges of deploying a Software-Defined Data Center (SDDC). Provide examples of scenarios where SDDC might be especially beneficial.

A Software-Defined Data Center (SDDC) is a data center where all infrastructure elements, including networking, storage, CPU, and security, are virtualized and delivered as a service. SDDC relies on software-based management to control the infrastructure, making it highly adaptable, scalable, and agile.

Benefits of SDDC:

Enhanced Agility: Allows rapid provisioning and deployment of resources, speeding up application development and reducing time-to-market.

Improved Scalability: Enables dynamic scaling of resources based on demand, supporting businesses that experience fluctuating workloads.

Operational Efficiency: Reduces manual configuration, automates processes, and centralizes management, leading to lower operational costs and fewer human errors.

Cost Savings: Reduces dependency on physical hardware, lowering capital expenses and minimizing data center footprint.

Enhanced Security and Compliance: Provides granular control over security policies and automates compliance, reducing risk.

Challenges of SDDC:

Complexity of Deployment: Initial setup and configuration can be complex and may require significant expertise.

Cost of Transition: Transitioning from traditional data centers to SDDC can involve significant costs, especially for businesses with legacy infrastructure.

Management of Legacy Systems: Integrating SDDC with existing legacy systems can be challenging and may limit the potential benefits of virtualization.

Security Risks: If not properly managed, SDDC environments can be vulnerable to security threats, as software-defined elements could be targeted by cyber attacks.

Scenarios Where SDDC is Especially Beneficial:

Cloud Service Providers: For companies offering cloud services, SDDC provides the flexibility, scalability, and automation needed to efficiently manage large-scale, multi-tenant environments.

Financial Services: In banking or insurance, SDDC can enhance data security, allow quick scaling, and support regulatory compliance through automated policy enforcement.

E-commerce Platforms: In retail, an SDDC can handle fluctuating traffic and resource demands, providing elasticity for peak shopping seasons and fast deployment of new services.

Research Institutions: Organizations involved in research often require high computational power and storage that can be dynamically allocated; SDDC provides this flexibility.

“Best of luck”